

Code & Data Execution Guide

To run the pipeline and replicate all reported metrics, please place all files in the same working directory and follow these execution paths:

1. Exploratory Data Analysis, Initial Preprocessing & CSV Generation

To evaluate the data-driven outlier removal process, view the correlation heatmap, inspect the exploratory baseline regression tests (including collinearity configurations), or review the initial logical grouping into album families alongside generating the final analysis, execute this notebook:

- Code Script to Open: `Taylor_swift_smlproject.py`
- Required Data Input: `taylor_swift_spotify.csv` (The raw dataset directly from the Spotify API)
- Outputs Generated: This script performs the initial data science lifecycle and exports the structured `taylor_baseline.csv` and `taylor_improved.csv` files used for the final models.

2. Running the Finalized Winning Models

To execute and evaluate the final optimized prediction architectures directly (Gradient Boosting, Random Forest Ensembles, and the stabilized OLS Linear Regression without energy), alongside generating the performance summary charts, execute this notebook:

- Code Script to Open: `winning_models_sml2.py`
- Required Data Inputs:
 - * `taylor_baseline.csv` (Tailored for OLS; contains 11 features)
 - * `taylor_improved.csv` (Tailored for Tree Ensembles; contains 12 features)